

Variance Risk Premium and Cross-Sectional Return in Chinese A-Share Markets

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May 2026

Abstract

We examine whether variance risk premium (VRP) exposure is priced in the cross-section of Chinese A-share returns. Using daily CSMAR data covering 4,777 unique stocks over 2015–2024, we construct VRP as the difference between a market-level implied variance derived from 50ETF options and a rolling Heterogeneous Auto-Regressive (HAR) forecast of realized variance. Embedding VRP betas in a Fama–MacBeth two-pass regression with controls for market beta, size, book-to-market, and momentum, we find a negative and statistically significant risk premium at daily and weekly frequencies, contrasting with the positive premia documented in U.S. markets. Full-sample estimates range from -0.85 to -0.96 with t -statistics exceeding 3.5. The premium is absent at the monthly horizon and concentrates sharply in high-volatility regimes, where estimates reach -2.74 ($t = -3.74$) at daily frequency. A nonparametric portfolio sort corroborates these findings: the high-minus-low VRP-beta spread averages -33.8 basis points per week, with a CH-4 four-factor alpha of -35.8 basis points, both statistically significant. We interpret the sign reversal as reflecting retail investor demand for high-volatility, lottery-like stocks, which bids up the prices of high-VRP-beta stocks and depresses their subsequent returns.

Contents

1	Introduction	3
2	Literature Review	3
3	VRP Construction	4
3.1	Definition	4
3.2	Physical-Measure Forecast: HAR Model	4
4	Data and Factor Construction	5
4.1	Data Sources	5
4.2	Sample Construction	5
4.3	Three-Factor Construction	5
4.4	Momentum Factor Construction	6
4.5	Factor Validation	6
4.6	Factor Correlation	7
5	Methodology	7
5.1	First Pass: Rolling Beta Estimation	8
5.2	Second Pass: Cross-Sectional Regression	8
5.3	Inference	8
5.4	Volatility Regime Classification	8

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6 Empirical Results	9
7 Robustness: Portfolio Sort Analysis	10
7.1 Portfolio Design	10
7.2 Results	10
7.3 Discussion	10
8 Conclusion	11
A Full Fama–MacBeth Regression Output	11
References	14

1 Introduction

The variance risk premium (VRP)—the spread between the risk-neutral and physical-measure expectations of future variance—has been established as a priced factor in developed equity markets. In the United States, stocks with higher VRP exposure earn higher subsequent returns, and the premium concentrates in periods of elevated market uncertainty [Han and Zhou, 2011, Bali, Engle, and Murray, 2016]. Whether the same mechanism operates in an emerging market dominated by retail investors and without a deep single-stock option market remains an open empirical question.

This paper answers that question for Chinese A-shares—and the answer inverts the U.S. evidence. We document a negative and statistically significant VRP risk premium at daily and weekly frequencies, with full-sample estimates between -0.85 and -0.96 ($t > 3.5$). The effect amplifies sharply in high-volatility regimes, reaching -2.74 ($t = -3.74$) at daily frequency, and is absent in calm markets. A nonparametric portfolio sort corroborates these findings without imposing linearity: the CH-4 alpha of the high-minus-low VRP-beta spread is -35.8 basis points per week, significant at the 1% level.

A key institutional constraint shapes our design. Throughout the sample 2015–2024, the 50ETF index option is the only liquid option in China; single-stock options are too thinly traded for reliable VRP construction. We therefore construct a market-level VRP series from 50ETF options using a rolling HAR model and estimate each stock’s sensitivity via rolling time-series regressions in the Fama–MacBeth first pass. We interpret the sign reversal as reflecting retail investor demand for high-volatility, lottery-like stocks, which bids up the prices of high-VRP-beta stocks and depresses their subsequent returns.

The remainder of the paper is organized as follows. Section 2 reviews the existing VRP literature. Section 3 describes VRP construction. Section 4 describes the data and factor construction. Section 5 presents the Fama–MacBeth framework. Section 6 reports empirical results. Section 7 presents the portfolio sort robustness check. Section 8 concludes.

2 Literature Review

A growing body of work examines whether VRP exposure is priced in the cross-section of stock returns. The existing evidence is almost exclusively from U.S. markets and relies on firm-level option data.

Han and Zhou (2011). Han and Zhou [2011] provide the first direct test of VRP as a cross-sectional pricing factor. Using individual equity options for U.S. stocks, they construct firm-level VRP as the difference between each stock’s implied and realized variance. In Fama–MacBeth regressions, they document a positive and highly significant risk premium ($t = 7.41$): stocks with greater exposure to variance risk earn higher subsequent returns. The premium concentrates in high-uncertainty periods, suggesting a state-dependent component to variance risk pricing.

Bali and Zhou (2016). Bali, Engle, and Murray [2016] embed VRP in a conditional ICAPM framework. They estimate each stock’s covariance with the aggregate VRP using a DCC-GARCH model and find a positive and significant uncertainty risk price. Their framework formalizes the variance-of-variance channel of Bollerslev, Tauchen, and Zhou [2009], whereby stocks that covary positively with rising variance uncertainty command a positive premium as compensation for bearing that risk.

Feunou, Jahan-Parvar, and Okou (2017). Feunou, Jahan-Parvar, and Okou [2017] show that total VRP, when left undecomposed, is insignificant in U.S. cross-sectional regressions ($t = 1.69$). Decomposing VRP into downside (“bad”) and upside (“good”) variance components restores significance ($t > 3$), with the downside component driving the result. Their finding cautions against treating total VRP as a homogeneous scalar.

Our contribution. All three studies share a common feature: firm-level VRP is constructed directly from individual equity options. This approach is not feasible in China, where single-stock options were introduced only in 2019 and remain thinly traded. Throughout our sample (2015–2024), the 50ETF index option is the only instrument with sufficient liquidity for reliable implied variance extraction. We therefore construct a

market-level VRP proxy and estimate each stock’s sensitivity to it via rolling time-series regressions in the FM first pass.

Our paper makes three contributions:

- **Sign reversal in an emerging market.** We document a negative and statistically significant VRP risk premium in Chinese A-shares—the opposite of what U.S. evidence predicts. Stocks with greater exposure to variance risk earn *lower* subsequent returns, suggesting that the standard compensation-for-risk narrative does not travel to retail-dominated markets.
- **Regime-dependent pricing with striking amplification.** The premium is essentially zero in calm markets but reaches -2.74 ($t = -3.74$) at daily frequency during high-volatility regimes—approximately 2.5 times the full-sample estimate. VRP pricing is a high-stress phenomenon, not a steady-state one.
- **Robust across methods.** A nonparametric weekly portfolio sort yields a CH-4 alpha of -35.8 basis points per week ($t = -2.77$), significant at the 1% level and larger in magnitude than the raw spread. The four-factor model not only fails to explain the return differential—it predicts the wrong sign.

3 VRP Construction

3.1 Definition

The variance risk premium at date t is defined as:

$$\text{VRP}_t = \text{IV}_{t,t+20} - \mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+20}], \quad (1)$$

where $\text{IV}_{t,t+20}$ is the 20-day implied variance extracted from 50ETF option prices, and $\mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+20}]$ is the physical-measure forecast of 20-day realized variance. A positive VRP reflects the insurance premium embedded in option prices above and beyond rational variance expectations.

The implied variance is computed from the China VIX index (annualized) as:

$$\text{IV}_{t,t+20} = \left(\frac{\text{VIX}_t}{100} \right)^2 \times \frac{20}{252}. \quad (2)$$

Daily realized variance is constructed from log-returns:

$$\text{RV}_t = r_t^2, \quad r_t = \ln \left(\frac{P_t}{P_{t-1}} \right), \quad (3)$$

and the 20-day cumulative realized variance is $\text{RV}_{t,t+20} = \sum_{\tau=t+1}^{t+20} \text{RV}_\tau$.

3.2 Physical-Measure Forecast: HAR Model

The physical expectation $\mathbb{E}_t^{\mathbb{P}}[\text{RV}_{t,t+20}]$ is unobservable and must be estimated from a statistical model. We use the Heterogeneous Auto-Regressive (HAR) model of Corsi [2009], which captures the long-memory structure of realized volatility by decomposing it into daily, weekly, and monthly components:

$$\text{RV}_{t,t+20} = \alpha + \beta_d \text{RV}_t^{(d)} + \beta_w \text{RV}_t^{(w)} + \beta_m \text{RV}_t^{(m)} + \varepsilon_t, \quad (4)$$

where the regressors are:

$$\text{RV}_t^{(d)} = \text{RV}_t, \quad \text{RV}_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 \text{RV}_{t-i}, \quad \text{RV}_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} \text{RV}_{t-i}. \quad (5)$$

The model is estimated in a strictly out-of-sample rolling fashion to avoid look-ahead bias. At each date t , equation (4) is estimated by OLS on the most recent 500 trading days ending at $t - 22$, and the

fitted coefficients are applied to the current regressors $(RV_t^{(d)}, RV_t^{(w)}, RV_t^{(m)})$ to obtain the one-shot 20-day forecast $\widehat{RV}_{t,t+20}^{\text{HAR}}$. The 22-day lag between the estimation window and the forecast date ensures that no future information enters the physical-measure estimate. Forecasts are floored at zero. Extreme observations of the raw VRP series beyond ± 6 standard deviations are set to missing before use in the cross-sectional regressions. The final series is scaled by 10,000 for numerical convenience.

4 Data and Factor Construction

4.1 Data Sources

Stock-level data are from CSMAR, covering all A-share stocks listed on the Shanghai and Shenzhen exchanges from January 2015 to December 2024 (9,250,096 daily observations before screening). Daily returns are measured using the price return excluding cash dividends (`Dretnd`), and total market capitalization (`Dsmvt11`) is used for value-weighting throughout. Balance sheet data (net assets, `a003100000`) are from annual reports (`typrep = A`). The risk-free rate is the three-month SHIBOR rate, provided at daily, weekly, and monthly frequencies.

4.2 Sample Construction

We follow the data-cleaning criteria of Shi [2023]. The following observations are excluded:

- Suspended stocks (`Trdsta` $\neq 1$);
- One-sided limit-hit days (`Hiprc = Loprc`);
- Risk-warning stocks (names containing “ST”, “*ST”, or “PT”);
- Delisted or delisting-risk stocks (`Statco = D`);
- Stocks listed for fewer than 12 months;
- Firm-years with non-positive net assets.

The final sample contains 4,777 unique stocks and 7,629,876 stock-day observations.

4.3 Three-Factor Construction

We construct MKT, SMB, and HML following Shi [2023], which adapts the Fama and French [1993] procedure to Chinese institutional features.

Market factor.

$$\text{MKT}_t = \frac{\sum_i \text{MV}_{i,t} \cdot r_{i,t}}{\sum_i \text{MV}_{i,t}}, \quad (6)$$

where $\text{MV}_{i,t}$ is total market capitalization and $r_{i,t}$ is the return of stock i at period t . At weekly and monthly frequencies, returns are compounded from daily observations as $r_{i,t} = \prod_{\tau \in t} (1 + r_{i,\tau}) - 1$.

Book-to-market ratio and rebalancing. In China, annual reports must be filed by end of April. Following Shi [2023], the rebalancing date is the last trading day of April and the matched fiscal year is:

$$y^* = \begin{cases} y - 1 & \text{if month}(t) \geq 5, \\ y - 2 & \text{if month}(t) \leq 4. \end{cases} \quad (7)$$

The book-to-market ratio $\text{BM}_{i,t} = \text{BookValue}_{i,y^*} / \text{MV}_{i,t}$ is winsorized at the 1st and 99th percentiles each month.

SMB and HML. Portfolios are formed via an independent 2×3 sort on size and BM. Breakpoints are computed using main-board stocks only (**Markettype** $\in \{1, 4\}$), so that small-cap ChiNext and STAR Market stocks do not distort the size breakpoint. The size breakpoint is the median market capitalization of main-board stocks; BM breakpoints are the 30th and 70th percentiles. All stocks are assigned to one of six value-weighted portfolios (S/L, S/M, S/H, B/L, B/M, B/H), held from May of year y through April of year $y + 1$:

$$\text{SMB}_t = \frac{1}{3} \left(r_t^{S/L} + r_t^{S/M} + r_t^{S/H} \right) - \frac{1}{3} \left(r_t^{B/L} + r_t^{B/M} + r_t^{B/H} \right), \quad (8)$$

$$\text{HML}_t = \frac{1}{2} \left(r_t^{S/H} + r_t^{B/H} \right) - \frac{1}{2} \left(r_t^{S/L} + r_t^{B/L} \right). \quad (9)$$

4.4 Momentum Factor Construction

We construct the momentum factor (UMD) following the classic algorithm of Shi [2023], which adapts the Carhart [1997] procedure to the Chinese market.

Momentum signal. The momentum variable for stock i at the end of month t is the cumulative return over the past 12 months excluding the most recent month. Following Shi [2023], one month is defined as 21 trading days. The signal is computed from the daily return series as:

$$\text{MOM}_{i,t} = \prod_{\tau=t-252}^{t-22} (1 + r_{i,\tau}) - 1, \quad (10)$$

where the window spans approximately 231 trading days. The signal is computed at each month-end using daily data. A minimum of 200 valid daily observations is required within the window; stocks with fewer observations are excluded.

Portfolio construction. At the end of each month t , all stocks are ranked by $\text{MOM}_{i,t}$. Breakpoints are computed using main-board stocks only (**Markettype** $\in \{1, 4\}$), consistent with the convention applied for SMB and HML. Stocks are divided into three groups at the 30th and 70th percentiles of the main-board momentum distribution. The UMD factor is:

$$\text{UMD}_t = \bar{r}_{t+1}^{\text{top } 30\%} - \bar{r}_{t+1}^{\text{bottom } 30\%}, \quad (11)$$

where $\bar{r}_{t+1}^{\text{top } 30\%}$ and $\bar{r}_{t+1}^{\text{bottom } 30\%}$ are the equal-weighted average returns of the high- and low-momentum portfolios in month $t + 1$. Unlike SMB and HML, the momentum factor uses equal-weighting following Shi [2023]. Portfolios are rebalanced monthly.

Momentum as a control characteristic. In the Fama–MacBeth cross-sectional regressions, $\text{MOM}_{i,t}$ enters directly as a firm-level characteristic in the second pass, analogously to SIZE and BM. It is not used to estimate a time-series beta in the first pass. This treatment follows Bali, Engle, and Murray [2016], who use characteristics rather than factor loadings for momentum.

4.5 Factor Validation

We validate our MKT, SMB, HML, and UMD series against the BetaPlus benchmark of Shi [2023] over the overlapping sample May 2016–December 2024. Table 1 reports time-series correlations at both monthly and daily frequencies.

Table 1: Factor Validation against BetaPlus Benchmark

Factor	Monthly Corr. ($T = 104$)	Daily Corr. ($T = 2,108$)
MKT	0.984	0.995
SMB	0.986	0.993
HML	0.973	0.976
UMD	0.880	0.875

Notes: Pearson time-series correlations between our constructed factors and the corresponding BetaPlus classic-algorithm series. Sample period: May 2016–December 2024. MKT, SMB, HML are constructed following the 2×3 sort described above. UMD is constructed using daily-based momentum signals and equal-weighted portfolios, rebalanced monthly.

Correlations for MKT, SMB, and HML exceed 0.97 at both frequencies. The UMD correlation is somewhat lower (0.880 monthly, 0.875 daily), which is expected: momentum factors are more sensitive to the exact window definition and rebalancing timing.

4.6 Factor Correlation

Figure 1 presents the pairwise correlation matrices of the five factors at monthly and daily frequencies. The structure is stable across both horizons. The strongest pairwise correlation is between MKT and HML (-0.38 at both frequencies), reflecting the tendency of growth stocks (low BM) to outperform in rising markets. SMB and HML are also negatively correlated (-0.34 monthly, -0.25 daily), consistent with the well-documented growth tilt of small-cap stocks in the Chinese market. HML and UMD exhibit moderate negative correlation (-0.26 monthly, -0.21 daily), in line with the value-momentum tension documented in Fama and French [1993] and subsequent literature. Importantly, VRP correlates weakly with all four equity factors (absolute correlations below 0.13 at both frequencies), indicating that VRP exposure captures information largely orthogonal to the standard size, value, and momentum characteristics. This low multicollinearity supports the validity of including VRP beta alongside SIZE, BM, and MOM as independent regressors in the Fama–MacBeth second pass.

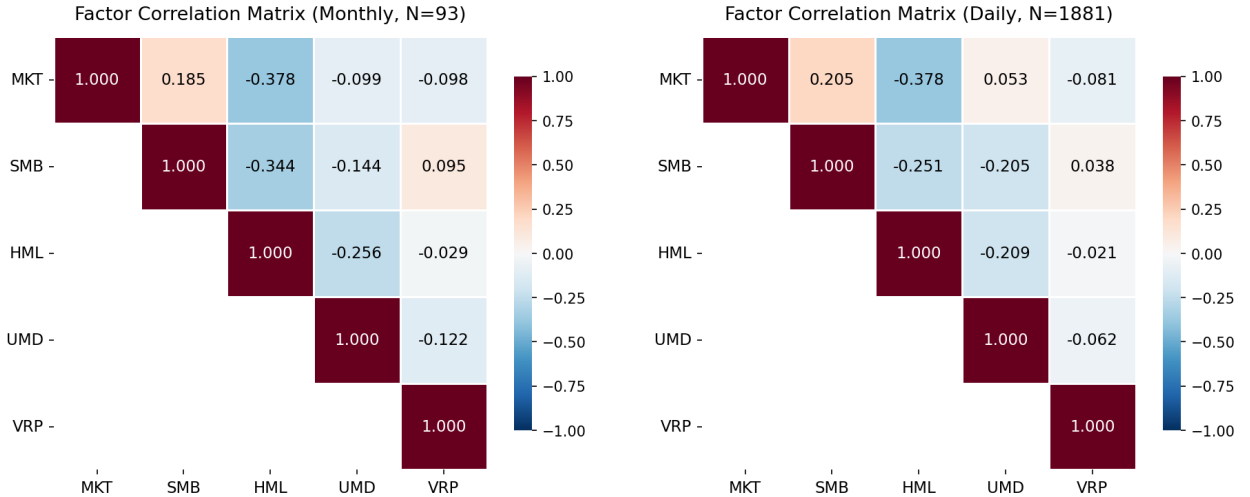


Figure 1: Factor Correlation Matrices. Left: monthly frequency ($T = 93$). Right: daily frequency ($T = 1,881$). Factors are MKT, SMB, HML (constructed following Shi 2023), UMD (momentum), and VRP (variance risk premium). Only the upper triangle is shown.

5 Methodology

We implement the Fama–MacBeth two-pass regression following Bali, Engle, and Murray [2016].

5.1 First Pass: Rolling Beta Estimation

For each stock i and each period t , factor exposures are estimated via a rolling time-series regression over the window $[t - W, t - 1]$:

$$r_{i,\tau}^e = \alpha_i + \beta_i^{\text{VRP}} \cdot \text{VRP}_\tau + \beta_i^{\text{MKT}} \cdot \text{MKT}_\tau^e + \varepsilon_{i,\tau}, \quad \tau \in [t - W, t - 1], \quad (12)$$

where $r_{i,\tau}^e = r_{i,\tau} - r_{f,\tau}$ is the excess return of stock i and $\text{MKT}_\tau^e = \text{MKT}_\tau - r_{f,\tau}$ is the market excess return. Rolling-window parameters $(W, W_{\min})$ are set to $(252, 120)$ at daily frequency, $(52, 26)$ at weekly frequency, and $(36, 24)$ at monthly frequency. Stocks with fewer than $W_{\min}$ valid observations in the estimation window are excluded from the cross-section for that period.

5.2 Second Pass: Cross-Sectional Regression

At each period t , we run an OLS cross-sectional regression of excess returns on the estimated betas and firm characteristics:

$$r_{i,t}^e = \lambda_{0,t} + \lambda_t^{\text{VRP}} \hat{\beta}_{i,t}^{\text{VRP}} + \lambda_t^{\text{MKT}} \hat{\beta}_{i,t}^{\text{MKT}} + \lambda_t^{\text{SIZE}} \text{SIZE}_{i,t} + \lambda_t^{\text{BM}} \text{BM}_{i,t} + \lambda_t^{\text{MOM}} \text{MOM}_{i,t} + \eta_{i,t}, \quad (13)$$

where $\text{SIZE}_{i,t} = \ln(\text{MV}_{i,t})$, $\text{BM}_{i,t}$ is the book-to-market ratio constructed as in Section 4, and $\text{MOM}_{i,t}$ is the momentum characteristic defined in equation (10). Following Bali, Engle, and Murray [2016], all independent variables are winsorized at the 0.5% level in each cross-section. The dependent variable is not winsorized. Cross-sections with fewer than 30 stocks are dropped.

5.3 Inference

The Fama–MacBeth risk premium estimate for VRP is:

$$\hat{\lambda}^{\text{VRP}} = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_t^{\text{VRP}}. \quad (14)$$

Statistical inference follows Newey and West [1987] with $L = 6$ lags. The Newey–West standard error is:

$$\widehat{\text{se}}_{\text{NW}} = \sqrt{\frac{1}{T} \left[\hat{\gamma}_0 + 2 \sum_{\ell=1}^L \left(1 - \frac{\ell}{L+1} \right) \hat{\gamma}_\ell \right]}, \quad (15)$$

where $\hat{\gamma}_\ell = \frac{1}{T} \sum_{t=\ell+1}^T (\hat{\lambda}_t^{\text{VRP}} - \hat{\lambda}^{\text{VRP}})(\hat{\lambda}_{t-\ell}^{\text{VRP}} - \hat{\lambda}^{\text{VRP}})$. The t -statistic is $t = \hat{\lambda}^{\text{VRP}} / \widehat{\text{se}}_{\text{NW}}$.

5.4 Volatility Regime Classification

To examine whether VRP pricing varies with market conditions, we partition the sample into three volatility regimes using a data-driven statistical classification. At each date t , we compute a rolling mean μ_t and standard deviation σ_t of the VIX index over the past 252 trading days (minimum 120 observations) and define:

$$\text{regime}_t = \begin{cases} \text{Low} & \text{if } \text{VIX}_t < \mu_t - \sigma_t, \\ \text{High} & \text{if } \text{VIX}_t > \mu_t + \sigma_t, \\ \text{Medium} & \text{otherwise.} \end{cases} \quad (16)$$

This rolling threshold adapts to the market’s own historical volatility baseline, avoiding the distortions that arise from fixed absolute cutoffs (e.g., $\text{VIX} > 25$) when the level of volatility shifts structurally over time. The regression in equation (13) is estimated separately for each regime subsample. The full-sample (“All”) results pool all periods regardless of regime.

6 Empirical Results

Table 2 reports the Fama–MacBeth risk premium estimates $\hat{\lambda}_{\text{VRP}}$ across three return frequencies, four volatility regime subsamples, and two control specifications. t -statistics in parentheses are based on Newey–West standard errors with six lags. Full regression output including all control variable coefficients is reported in Appendix Tables 4–9.

Table 2: Fama–MacBeth VRP Risk Premium Estimates

Frequency	Spec	Volatility Regime			
		All	Low VIX	Medium VIX	High VIX
Daily	No MOM	−0.8960*** (−3.74)	−0.4041 (−0.90)	−0.5813 (−1.35)	−2.2836** (−2.36)
	MOM	−0.9551*** (−3.74)	−0.0665 (−0.14)	−0.6548 (−1.47)	−2.7437*** (−2.61)
Weekly	No MOM	−0.8536** (−3.55)	−0.1882 (−0.51)	−0.6195 (−1.41)	−2.2804** (−2.51)
	MOM	−0.9121*** (−3.54)	−0.0527 (−0.14)	−0.6763 (−1.58)	−2.5711** (−2.35)
Monthly	No MOM	0.3314 (0.75)	0.2393 (0.40)	0.0989 (0.12)	2.3742 (0.75)
	MOM	−0.2584 (−0.68)	−0.0985 (−0.24)	−0.3070 (−0.42)	−0.3397 (−0.11)
T (daily)		1761	408	985	368
T (weekly)		370	88	207	75
T (monthly)		69	17	46	6

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

No MOM: controls are market beta, SIZE, BM. MOM: additionally controls for momentum.

†Monthly High VIX based on $T = 6$ periods; inference unreliable.

Three patterns emerge consistently across both specifications.

Negative risk premium. The estimated $\hat{\lambda}_{\text{VRP}}$ is negative at the daily and weekly frequencies across all regime subsamples, reversing the positive premium documented for U.S. stocks by Han and Zhou [2011] and Bali, Engle, and Murray [2016]. At the monthly frequency, estimates are uniformly insignificant and unstable in sign, suggesting that any monthly-horizon signal is too weak to be reliably identified in the cross-section. The consistently negative premium at shorter horizons is our central finding: whereas U.S. stocks with higher VRP exposure earn higher subsequent returns as compensation for bearing variance uncertainty, A-share investors appear to accept lower returns for stocks that hedge against variance risk. We defer a structural interpretation to Section 8, but note that this pattern is consistent with the dominance of retail investors who may overweight lottery-like payoffs associated with high-volatility stocks, bidding up their prices and depressing subsequent returns.

Significance across frequencies. At the daily and weekly frequencies, the full-sample VRP risk premium is statistically significant at the 1% level in both specifications. In the baseline (no momentum) specification, $\hat{\lambda}_{\text{VRP}}$ equals -0.896 ($t = -3.74$) at daily frequency and -0.854 ($t = -3.55$) at weekly frequency. The augmented specification adds momentum as an additional control following Bali, Engle, and Murray [2016]; the estimates shift modestly to -0.955 ($t = -3.74$) and -0.912 ($t = -3.54$), with t -statistics essentially unchanged. This insensitivity to momentum control indicates that the VRP effect is not subsumed by return-continuation patterns. At the monthly frequency, all estimates are statistically insignificant regardless of

specification or regime, consistent with the hypothesis that VRP information is rapidly incorporated into A-share prices given the high turnover and short holding horizons that characterise retail-dominated markets.

High-VIX amplification. The VRP risk premium concentrates sharply in high-volatility regimes. In the baseline specification, the high-VIX subsample yields $\hat{\lambda}_{\text{VRP}} = -2.28^{**}$ at both daily and weekly frequencies—approximately 2.5 times larger in magnitude than the full-sample estimates. The high-VIX estimates are similarly large in the augmented specification (-2.74^{***} daily, -2.57^{**} weekly), confirming that the amplification is not driven by momentum patterns in turbulent periods. By contrast, the low- and medium-VIX subsamples produce uniformly insignificant estimates across all frequencies and specifications, suggesting that VRP pricing is essentially inactive during calm market conditions. This state-dependent pattern is consistent with Bollerslev, Tauchen, and Zhou [2009], who document that the aggregate VRP is most informative about risk premia during periods of elevated uncertainty.

7 Robustness: Portfolio Sort Analysis

To complement the parametric Fama–MacBeth evidence, we conduct a nonparametric portfolio sort test that does not impose linearity on the relationship between VRP exposure and subsequent returns.

7.1 Portfolio Design

At the end of each week t , we assign each stock to one of ten decile portfolios based on its pre-formation $\hat{\beta}_{i,t}^{\text{VRP}}$, estimated from the same rolling time-series regression used in the FM first pass (equation (12), window $W = 52$ weeks, minimum $W_{\min} = 26$ observations). Decile boundaries are recomputed cross-sectionally each week; stocks with insufficient estimation history are excluded. Within each decile, portfolio returns are value-weighted using end-of-week- t market capitalisation as weights, and the holding period is the subsequent week $t+1$. This ensures strict out-of-sample formation: no information from week $t+1$ enters the sorting variable.

The primary test asset is the long–short spread Q10 – Q1: long the decile with the highest VRP beta and short the decile with the lowest. We report two statistics. First, the raw value-weighted excess return of the spread, with a Newey–West t -statistic using six lags. Second, the intercept from a time-series regression of the spread on the CH-4 factors (MKT, SMB, HML, UMD), which measures the return unexplained by standard risk exposures.

7.2 Results

Table 3: Q10–Q1 Long–Short Spread: Raw Return and CH-4 Alpha

	Mean (bps/week)	NW t -stat	p -value	Sig.	Ann. Sharpe / IR
Raw spread	−33.76	−2.41	0.017	**	−0.89
CH-4 alpha	−35.83	−2.77	0.006	***	−0.99

$n = 361$ weeks. Newey–West standard errors with 6 lags.

CH-4 factors: MKT, SMB, HML, UMD. Value-weighted portfolios.

Table 3 reports the results. The raw spread averages −33.8 basis points per week, significant at the 5% ($t = -2.41$, $p = 0.017$), with an annualized Sharpe ratio of −0.89. After controlling for the four CH-4 risk factors, the alpha is −35.8 basis points per week, significant at the 1% ($t = -2.77$, $p = 0.006$), with an annualized information ratio of −0.99.

7.3 Discussion

Two features of these results are notable. First, the direction of the spread is consistent with the FM evidence: stocks with the highest VRP beta exposure earn lower subsequent returns than those with the

lowest exposure, confirming the negative risk premium in a nonparametric setting that does not rely on linearity or distributional assumptions.

Second, the CH-4 alpha is more negative than the raw spread. This indicates that the standard four factors not only fail to explain the spread, but that factor-neutral returns are slightly larger in magnitude. In other words, the four-factor model assigns a positive expected return to the long-short position based on its factor loadings, which makes the risk-adjusted underperformance of high- β_{VRP} stocks even starker. This pattern is consistent with the FM finding that VRP beta captures information orthogonal to MKT, SMB, HML, and UMD exposures, as documented in the low pairwise correlations reported in Section 4.

Taken together, the portfolio sort results corroborate the FM evidence along two dimensions: the sign of the VRP risk premium is robustly negative, and the effect is not subsumed by standard four-factor risk adjustments.

8 Conclusion

This paper provides the first cross-sectional test of variance risk premium pricing in Chinese A-shares. Using daily CSMAR data covering 4,777 stocks over 2015–2024, we construct a market-level VRP from 50ETF options and a rolling HAR forecast of realized variance, estimate each stock’s VRP exposure via rolling time-series regressions, and embed these betas in a Fama–MacBeth two-pass framework with controls for market beta, size, book-to-market, and momentum.

Three findings stand out. First, the VRP risk premium is negative and statistically significant at daily and weekly frequencies, reversing the positive premium documented in U.S. markets by Han and Zhou [2011] and Bali, Engle, and Murray [2016]. Full-sample estimates range from -0.85 to -0.96 across specifications, with t -statistics exceeding 3.5 in absolute value. At the monthly frequency, estimates are uniformly insignificant, suggesting that VRP information is incorporated into prices within days to weeks rather than persisting to a monthly horizon. Second, the premium concentrates sharply in high-volatility regimes: high-VIX subsample estimates are approximately 2.5 times larger in magnitude than full-sample estimates and reach -2.74 at daily frequency, significant at the 1% level. In low- and medium-VIX regimes, pricing is essentially absent. Third, a nonparametric portfolio sort corroborates the parametric evidence: the long-short spread between the highest and lowest VRP-beta deciles averages -33.8 basis points per week (-35.8 bps after CH-4 risk adjustment), significant at the 5% and 1% levels respectively, and the four-factor alpha exceeds the raw spread in magnitude, indicating that standard risk exposures do not account for the differential.

The sign reversal relative to U.S. evidence is the central puzzle raised by our findings. We offer a tentative interpretation rooted in the structural features of the A-share market. Retail investors, who account for the majority of A-share trading volume, tend to exhibit a preference for high-volatility, lottery-like stocks. This demand drives up the prices of stocks with high VRP beta—those most sensitive to aggregate variance risk—depressing their subsequent returns relative to low-beta stocks. Under this mechanism, the VRP beta proxies for a lottery characteristic rather than a compensation for bearing undesirable risk, reversing the sign of the premium relative to institutional markets where variance risk is genuinely undesirable to the marginal investor. The state-dependence of the effect—stronger in high-VIX regimes—is consistent with this interpretation: retail demand for lottery stocks intensifies precisely when aggregate uncertainty is elevated and the variance risk premium is large.

Several limitations of our analysis merit acknowledgment. The use of a market-level rather than firm-level VRP is a data-driven constraint rather than a modelling choice, and the rolling beta estimates conflate true VRP sensitivity with measurement error. The structural interpretation offered above is speculative and would benefit from direct tests using investor composition data. Future work could examine whether the sign reversal persists across different sub-periods, extend the analysis to other emerging markets with similar retail dominance, and explore whether decomposing the VRP into downside and upside components in the spirit of Feunou, Jahan-Parvar, and Okou [2017] sharpens identification in the Chinese context.

A Full Fama–MacBeth Regression Output

Tables 4–9 report the complete Fama–MacBeth regression output for all control variables across the baseline (3-factor, no momentum) and augmented (4-factor with momentum) specifications. t -statistics in parentheses

are based on Newey–West standard errors with six lags.

Table 4: Full FM Output, Daily Frequency (3-Factor, No Momentum)

Factor T	All 1761	Low VIX 408	Medium VIX 985	High VIX 368
VRP	-0.8960^{***} (-3.74)	-0.4041 (-0.90)	-0.5813 (-1.35)	-2.2836^{**} (-2.36)
MKT	-1.78×10^{-4} (-0.91)	-2.60×10^{-5} (-0.05)	-3.24×10^{-4} (-0.91)	4.27×10^{-5} (0.06)
SIZE	$4.33 \times 10^{-4}^{***}$ (6.17)	$4.64 \times 10^{-4}^{***}$ (3.20)	$4.32 \times 10^{-4}^{***}$ (3.67)	3.98×10^{-4} (1.23)
BM	$-1.77 \times 10^{-6}^{***}$ (-13.18)	$-1.38 \times 10^{-6}^{***}$ (-4.12)	$-1.60 \times 10^{-6}^{***}$ (-7.08)	$-2.38 \times 10^{-6}^{***}$ (-4.53)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM. VRP = $IV_t - RV_t$.

Table 5: Full FM Output, Weekly Frequency (3-Factor, No Momentum)

Factor T	All 370	Low VIX 88	Medium VIX 207	High VIX 75
VRP	-0.8536^{**} (-3.55)	-0.1882 (-0.51)	-0.6195 (-1.41)	-2.2804^{**} (-2.51)
MKT	-4.75×10^{-4} (-0.84)	-9.32×10^{-4} (-0.81)	-0.0020^{**} (-2.34)	0.0043 (1.65)
SIZE	0.0021^{***} (6.08)	0.0021^{***} (3.06)	0.0019^{***} (3.27)	0.0025 (1.57)
BM	$-8.51 \times 10^{-6}^{***}$ (-11.12)	$-8.42 \times 10^{-6}^{***}$ (-4.37)	$-7.87 \times 10^{-6}^{***}$ (-5.83)	$-1.13 \times 10^{-5}^{***}$ (-4.11)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM. VRP = $IV_t - RV_t$.

Table 6: Full FM Output, Monthly Frequency (3-Factor, No Momentum)

Factor T	All 69	Low VIX 17	Medium VIX 46	High VIX [†] 6
VRP	0.3314 (0.75)	0.2393 (0.40)	0.0989 (0.12)	2.3742 (0.75)
MKT	-0.0057^{***} (-4.80)	-0.0079^{***} (-4.04)	-0.0045^{**} (-2.13)	-0.0088 (-1.41)
SIZE	0.0067^{***} (5.13)	0.0028 (1.09)	0.0090^{***} (3.56)	0.0009 (0.35)
BM	$-3.29 \times 10^{-5}^{***}$ (-9.68)	$-1.88 \times 10^{-5}^{***}$ (-4.12)	$-3.60 \times 10^{-5}^{***}$ (-6.06)	$-4.87 \times 10^{-5}^{***}$ (-4.72)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM. VRP = $IV_t - RV_t$.

[†]Based on only $T = 6$ periods; Newey–West inference unreliable.

Table 7: Full FM Output, Daily Frequency (4-Factor with Momentum)

Factor T	All 1761	Low VIX 408	Medium VIX 985	High VIX 368
VRP	-0.9551^{***} (-3.74)	-0.0665 (-0.14)	-0.6548 (-1.47)	-2.7437^{***} (-2.61)
MKT	-2.74×10^{-4} (-1.42)	-5.20×10^{-5} (-0.10)	-3.92×10^{-4} (-1.11)	-2.05×10^{-4} (-0.29)
SIZE	$4.82 \times 10^{-4***}$ (7.19)	$4.61 \times 10^{-4***}$ (3.27)	$4.52 \times 10^{-4***}$ (4.11)	$5.87 \times 10^{-4*}$ (1.87)
BM	$-2.02 \times 10^{-6***}$ (-15.87)	$-1.47 \times 10^{-6***}$ (-4.80)	$-1.76 \times 10^{-6***}$ (-8.34)	$-3.06 \times 10^{-6***}$ (-5.92)
MOM	$-9.19 \times 10^{-4***}$ (-6.10)	-4.41×10^{-4} (-1.15)	$-6.95 \times 10^{-4***}$ (-2.75)	$-2.05 \times 10^{-3***}$ (-3.47)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM, MOM. VRP = $IV_t - RV_t$.

Table 8: Full FM Output, Weekly Frequency (4-Factor with Momentum)

Factor T	All 370	Low VIX 88	Medium VIX 207	High VIX 75
VRP	-0.9121^{***} (-3.54)	-0.0527 (-0.14)	-0.6763 (-1.58)	-2.5711^{**} (-2.35)
MKT	$-9.43 \times 10^{-4*}$ (-1.71)	-0.0011 (-0.94)	-0.0024^{***} (-2.78)	0.0032 (1.27)
SIZE	0.0024^{***} (7.47)	0.0021^{***} (3.27)	0.0022^{***} (3.87)	0.0034^{**} (2.36)
BM	$-9.97 \times 10^{-6***}$ (-13.25)	$-8.35 \times 10^{-6***}$ (-5.85)	$-8.68 \times 10^{-6***}$ (-6.94)	$-1.43 \times 10^{-5***}$ (-5.13)
MOM	-0.0053^{***} (-6.79)	-0.0028 (-1.30)	-0.0047^{***} (-3.90)	-0.0101^{***} (-3.08)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM, MOM. VRP = $IV_t - RV_t$.

Table 9: Full FM Output, Monthly Frequency (4-Factor with Momentum)

Factor T	All 69	Low VIX 17	Medium VIX 46	High VIX [†] 6
VRP	−0.2584 (−0.68)	−0.0985 (−0.24)	−0.3070 (−0.42)	−0.3397 (−0.11)
MKT	−0.0057*** (−3.82)	−0.0099*** (−3.81)	−0.0042 (−1.63)	−0.0055 (−1.25)
SIZE	0.0074*** (5.51)	0.0038 (1.31)	0.0091*** (3.47)	0.0042 (1.79)
BM	-3.46×10^{-5} *** (−10.51)	-2.27×10^{-5} *** (−6.41)	-3.63×10^{-5} *** (−6.87)	-6.22×10^{-5} *** (−6.33)
MOM	−0.0166*** (−5.23)	−0.0249** (−2.88)	−0.0109*** (−3.41)	−0.0373** (−3.33)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Newey–West t -statistics with 6 lags in parentheses.

Controls: market beta, SIZE, BM, MOM. VRP = $IV_t - RV_t$.

[†]Based on only $T = 6$ periods; Newey–West inference unreliable.

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